

Homework 4

You have a stove with an electric heating element that runs 10 A of current with 120 V potential difference. You have to heat up 1 kg of water from 30 °C to 100 °C. How long will it take? Assume that there is no thermal loss in heat conduction from the pot to the water, or from the water to the air.

1. Calculate the power dissipation from the heating element. Your answer should be in Watts. (2 points)
2. Calculate how much heat you need to raise the water temperature from 30 °C to 100 °C. Your answer should be in Joules. (2 points)
 - a. Hint: The heat capacity of water is 4,184 J/(kg·°C).
3. Calculate how long it will take. Your answer should be in seconds. (2 points)